Feedback on F79MA December 2024 Exam

Question 1:
Few students achieved near full marks on this question, with the majority scoring below average.

Specific Comments on Each Part of Q1:

1(a)(i) Most students correctly applied the Method of Moments Estimator (MME) to find the estimate, but many struggled with the Maximum Likelihood Estimator (MLE).
(ii) While most students successfully found the answers for the mean and variance of the MME estimator, they found it challenging to do the same for the MLE estimator. However, the concept of unbiased and biased estimators was generally well understood.
(iii) The majority of students were able to derive the Mean Squared Error (MSE) for the MME estimator, but only a small group managed to do so for the MLE estimator. Most students, however, provided correct justifications for selecting the better estimator.
(iv) Although most students understood how to find the confidence interval, many made mistakes midway through the process or struggled with correctly calculating the variance of the distribution.

1(b) Over half of students were unable to show that T is a sufficient statistic for the parameter θ. Many provided partial answers and failed to mention that T is a one-to-one function of the sufficient statistic identified from the likelihood function.

Question 2:

Students generally did fairly well on Question 2.

Q2(a) Some students struggled with writing down the likelihood function correctly, taking logs or taking derivatives, but most did okay. As a "show" type question, to get full marks a reasonable number of steps had to be shown to get full marks.

Q2(b) Generally answered well.

Q2(c) This was generally well answered but some students believed the score function could be rewritten in the form of the corollary.

Q2(d) Some students wrote the information function as minus the information function and some wrote down negative variances. Note that the Fisher information is an expected value of a random variable which in this case is the same as the observed Fisher information.

Q2.(e) There were multiple ways of answering this question so any correct solution was able to receive full marks.

Question 3:

Q3(a) Fewer than a quarter of students had been able to derive the likelihood ratio for a single observation x. Instead, the majority of students derived the likelihood ratio for a sample (x1, ..., xn) of size n for this part. Very few students actually explained how the threshold k' could be derived as a quantile of the chi-squared distribution with 1 degree of freedom. There were even fewer students who stated the values of x that favour H0, which was required as part of the question.

Q3(b) Most students had been able to derive the likelihood ratio correctly for this part. However, most students lost marks for not simplifying the likelihood ratio, not specifying the critical region correctly, and/or not explaining how the threshold k' could be derived as a quantile of the chi-squared distribution with n degrees of freedom.

Q3(c) As four marks are allocated to this question, students are expected to justify that the likelihood ratio is monotonically decreasing in the statistic sum_{x_i^2}. Only very few students had managed to give a complete justification. Among students who had chosen the "exponential family" approach, many did not express the individual probability density function of X in an appropriate format, which caused them to lose marks.

Q3(d) This part has a surprisingly low number of correct attempts. Only roughly 10% of the students were able to correctly identify the threshold of the rejection region and compare it against the data / calculate a p-value of the test. It is suggested that some exercises as such are added to the tutorials for the chapter "Hypothesis Testing".

Question 4:

Q4(a) The majority of the students derived the posterior distribution correctly. However, a handful of students failed to mention (explicitly) that the Bayes estimator for theta is the posterior mean.

Q4(b) Fewer than one third of students were able to correctly express the posterior mean in the credibility form. The majority of students were able to deduce (or guess) that mu = alpha/beta = prior mean, however only a few students were able to identify the values of Z and T. There were some attempts in using the Bulhmann's formula to derive the credibility factor Z, but none of them were successful. Very few students provided the interpretation to the statistic T, which was required as part of the question.

Q4(c) Students who did not solve part (b) correctly generally got very little marks for this part. (i) The majority of students only stated that Z increases, but not that Z increases to 1. (ii) The majority of students were able to deduce that mu remains unchanged as n increases. (iii) Only a few students correctly stated that the Bayes estimator converges to T.

Question 5:

This problem was generally done well by students, since it is similar to problems seen in the tutorials and past exam papers.